

# Jiawei ZHANG

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## EDUCATION

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### University of Chicago

Ph.D. in Computer Science, 4.0/4.0

Advisor: *Prof. Bo Li*

Sept. 2024 – Aug. 2026

### University of Illinois Urbana-Champaign

Ph.D. in Computer Science, 4.0/4.0

Advisor: *Prof. Bo Li*

Aug. 2023 – May. 2024

M.S. in Computer Science

May. 2023

### Zhejiang University

Bachelor of Engineering

Hangzhou, China

Sept. 2017 – Jun. 2021

## RESEARCH INTEREST:

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I study **Safe AGI**, with a focus on **making LLMs and LLM agents reliable at scale**. My research runs as a closed loop: **(i) scalable red-teaming** to elicit realistic, long-horizon failures. I participated in internal red-teaming evaluations of **OpenAI o1**, **Google DeepMind**, and **ElevenLabs TTS** with **Virtue AI**. In parallel, I study **agentic reliability** for LLM agents, focusing on agent-specific vulnerabilities (e.g., prompt injection and malicious action execution) as well as **environment security construction** for agentic systems; **(ii) defense & alignment** that transforms safety requirements into actionable safeguards, spanning **certified robustness** for worst-case guarantees and **interpretability-guided defenses** that translate mechanistic insights into generalizable mitigations; and **(iii) better safety-aware reasoning for autonomous driving** by incorporating richer world knowledge via reinforcement learning for vision–language–action models.

Together, I aim to build **scalable red-teaming pipelines for LLM agents** and develop **practical patches** that improve safety awareness **while maintaining scalability**. Previously, I was an **Anthropic Fellow for AI Safety Research** from January to July 2026.

## SELECTED PUBLICATION (\* DENOTES CO-FIRST AUTHORSHIP)

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- **Jiawei Zhang**, Andrew Estornell, David D. Baek, Bo Li, Xiaojun Xu. Any-Depth Alignment: Unlocking Innate Safety Alignment of LLMs to Any-Depth. *International Conference on Learning Representations (ICLR) 2026*. [arXiv]
- Zhaorun Chen\*, Xun Liu\*, Mintong Kang, **Jiawei Zhang**, Minzhou Pan, Shuang Yang, Bo Li. ARMs: Adaptive Red-Teaming Agent against Multimodal Models with Plug-and-Play Attacks. *International Conference on Learning Representations (ICLR) 2026*. [arXiv]
- Mintong Kang\*, Zhaorun Chen\*, Chejian Xu\*, **Jiawei Zhang\***, Chengquan Guo\*, Minzhou Pan, Ivan Revilla, Yu Sun, Bo Li. GuardSet-X: Massive Multi-Domain Safety Policy-Grounded Guardrail Dataset. *NeurIPS 2025*. [arXiv]
- **Jiawei Zhang**, Shuang Yang, Bo Li. UDora: A Unified Red Teaming Framework Against LLM Agents by Dynamically Leveraging Their Own Reasoning. *International Conference on Machine Learning (ICML) 2025*. [arXiv]
- **Jiawei Zhang**, Xuan Yang, Taiqi Wang, Yu Yao, Aleksandr Petiushko, Bo Li. SafeAuto: Knowledge-Enhanced Safe Autonomous Driving with Multimodal Foundation Models. *International Conference on Machine Learning (ICML) 2025*. [arXiv]
- Chejian Xu\*, **Jiawei Zhang\***, Zhaorun Chen\*, Chulin Xie\*, Mintong Kang\*, Zhuowen Yuan\*, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, ..., Zidi Xiong, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li. MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models. *International Conference on Learning Representations (ICLR) 2025*. [arXiv]
- **Jiawei Zhang**, Chejian Xu, Yu Gai, Freddy Lecue, Dawn Song, Bo Li. KnowHalu: Hallucination Detection via Multi-Form Knowledge Based Factual Checking. *ICLR 2025 Workshop on Foundation Models in the Wild*. [arXiv]
- Bowen Jin, Chulin Xie, **Jiawei Zhang**, Kashob Kumar Roy, Yu Zhang, Zheng Li, Ruirui Li, Xianfeng Tang, Suhang Wang, Yu Meng, Jiawei Han. Graph Chain-of-Thought: Augmenting Large Language Models by Reasoning on Graphs. *Findings of the Association for Computational Linguistics (ACL) 2024*. [arXiv]

- **Jiawei Zhang**, Chejian Xu, Bo Li. ChatScene: Knowledge-Enabled Safety-Critical Scenario Generation for Autonomous Vehicles. *Conference on Computer Vision and Pattern Recognition (CVPR) 2024*. [arXiv]
- **Jiawei Zhang**, Tianyu Pang, Chao Du, Yi Ren, Bo Li, Min Lin. MMCBench: Benchmarking Large Multimodal Models against Common Corruptions. [arXiv]
- **Jiawei Zhang**, Zhongzhu Chen, Huan Zhang, Chaowei Xiao, Bo Li. DiffSmooth: Certifiably Robust Learning via Diffusion Models and Local Smoothing. *32th USENIX Security Symposium 2023*. [arXiv]
- **Jiawei Zhang**, Linyi Li, Ce Zhang, Bo Li. CARE: Certifiably Robust Learning with Reasoning via Variational Inference. *IEEE Conference on Secure and Trustworthy Machine Learning (SatML) 2023*. [arXiv]
- Zhuolin Yang\*, Zhikuan Zhao\*, Boxin Wang, **Jiawei Zhang**, Linyi Li, Hengzhi Pei, Bojan Karlas, Ji Liu, Heng Guo, Ce Zhang, Bo Li. Improving Certified Robustness via Statistical Learning with Logical Reasoning. *Advances in Neural Information Processing Systems (NeurIPS) 2022*. [arXiv]
- Linyi Li, **Jiawei Zhang**, Tao Xie, Bo Li. Double Sampling Randomized Smoothing. *International Conference on Machine Learning (ICML) 2022*. [arXiv]
- **Jiawei Zhang\***, Linyi Li\*, Huichen Li, Xiaolu Zhang, Shuang Yang, Bo Li. Progressive-Scale Boundary Blackbox Attack via Projective Gradient Estimation. *International Conference on Machine Learning (ICML) 2021*. [arXiv]

## INDUSTRY RESEARCH EXPERIENCE

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### OpenAI

*Safety Research, Member of Technical Staff*

*Aug 2026 – Present*

### Anthropic

*AI Safety Research Fellow*

*Jan 2026 – July 2026*

- Automatic generation of environments and attacks for prompt injection evaluation at scale.

### NVIDIA

*ML Research Intern, Santa Clara*

*Advised by Dr. Boris Ivanovic and Prof. Marco Pavone*

*Oct 2025 – Jan 2026*

- Explored improved reinforcement learning methods for reasoning to reduce minADE.
- Developed enhanced reasoning capabilities for autonomous driving by leveraging world models.

### ByteDance Seed Research

*Responsible AI Research Intern, San Jose*

*Advised by Dr. Xiaojun Xu and Dr. Hang Li*

*June 2025 – Sept 2025*

- Developed a scalable, model-agnostic defense for LLMs based on safety-signature tokens, effective against deep prefill attacks with context lengths exceeding 3,000 tokens.
- Demonstrated robustness to in-the-wild jailbreaks (AutoDAN, PAIR) while maintaining 0% false positive rate on standard benchmarks (MMLU, MATH, BBH, AlphaEval).

### Nuro AI

*Machine Learning Research Intern (Pathfinder), Mountain View*

*Advised by Dr. Aleksandr Petiushko*

*May 2024 – Aug 2024*

- Fine-tuned a multimodal LLM for video-conditioned autonomous driving to output high-level action plans and low-level control signals, unifying perception, reasoning, and control.
- Developed a multimodal retrieval-augmented training pipeline and a Markov Logic Network that encodes first-order traffic rules to verify and improve the safety of LLM-proposed actions.
- Proposed a position-dependent cross-entropy loss (PDCE) to stabilize and improve numeric control predictions when represented as text.

### Sea AI Lab

*Machine Learning Research Intern, Singapore*

*Advised by Dr. Tianyu Pang & Dr. Chao Du*

*May 2023 – Aug 2023*

- Conducted evaluations on cross-modal models (e.g., Stable Diffusion, Whisper) to assess their consistency under a range of common data corruptions.
- Developed a rigorous benchmark for assessing the self-consistency of these models. The benchmark was designed to provide a comprehensive understanding of model behavior, incorporating a wide range of scenarios and inputs to measure their resilience and accuracy.

## WORKSHOPS AND COMPETITIONS

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- **CLAS 2024: The Competition for LLM and Agent Safety, NeurIPS 2024**
- **Secure and Safe Autonomous Driving (SSAD) Workshop and Challenge, CVPR 2023**

## TEACHING

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### **CS 307 - Modeling and Learning in Data Science (Spring 2022)**

- Teaching Assistant with *Prof.* Bo Li and *Prof.* David Forsyth

## SELECTED HONOR & AWARDS

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|-------------------------------------------------------------------------------------------|--------------------|
| <b>Graduate Conference Travel Award</b>                                                   | <i>March. 2023</i> |
| <b>Accumulative Research Innovation &amp; Entrepreneurship Index: 1st Department-wide</b> | <i>June. 2021</i>  |
| <b>Meritorious Winner, MCM COMAP's Mathematical Contest in Modeling</b>                   | <i>Apr. 2020</i>   |
| <b>First Prize, China Harbour Scholarship</b>                                             | <i>Jan. 2020</i>   |
| <b>First Prize, The Chinese Mathematics Competitions, Zhejiang Province</b>               | <i>Nov. 2018</i>   |

## PROFESSIONAL SERVICE

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**Session Chair** — ICLR 2025

**Top Reviewer** — NeurIPS 2025

**Program Committee**— ICML, NeurIPS, ICLR, CVPR, ECCV, AISTATS, AAAI, COLM, ACL, JIMR, etc.